Quantum Teleportation (I)

Background: What and why?

Pinakin Padalia

What are we covering in this module?

- Quantum Teleportation: What, why and how?
- Superdense coding: What, why and how?
- Quantum simulations: QpiAI explorer examples
- Quantum Cryptography and QKD
- Quantum Communication and Networks

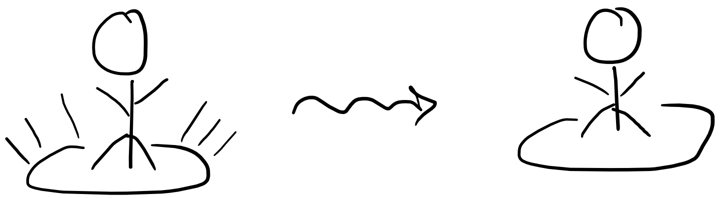
What are we covering in this chapter?

- **Quantum Teleportation: What and why?**
  - *History and big picture*
  - *Why teleportation*
  - *Difference between quantum and classical*
  - *Applications*
- Quantum Teleportation: How?
- QpiAI Explorer demonstration

Teleportation: History

"Beam me Up!"

- Theoretical demonstration: 1993 [Bennet et. al.]
- Practical demonstration: 1997 [Bouwmeester et. al.]



"Telepheresis"

PHYSICAL REVIEW LETTERS

VOLUME 70 | 29 MARCH 1993 | NUMBER 13

Teleporting an Unknown Quantum State via Dual Classical and Einstein-Podolsky-Rosen Channels

Charles H. Bennett,(1) Gilles Brassard,(2) Claude Crépeau,(2),(3) Richard Jozsa,(2) Asher Peres,(4) and William K. Wootters(5)

Published: 01 December 1997

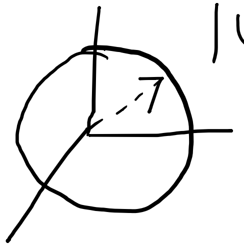
Experimental quantum teleportation

Dik Bouwmeester, Jian-Wei Pan, Klaus Mattle, Manfred Eibl, Harald Weinfurter & Anton Zeilinger

Nature **390**, 575–579 (1997) | Cite this article

Teleportation: What is it?

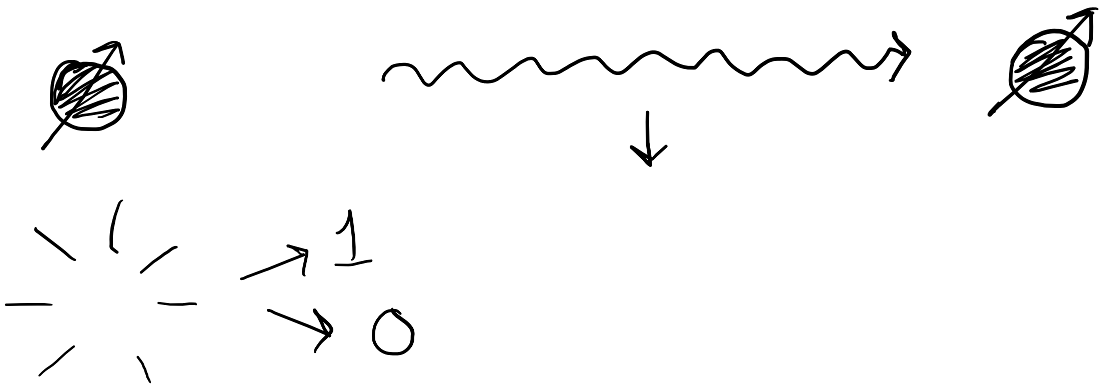
- What are we planning to teleport?
- Information in a qubit: complex coefficients: $|\Psi> = \alpha|0> + \beta|1>$
- Essentially transfer of this information



$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$\alpha, \beta \in C$$

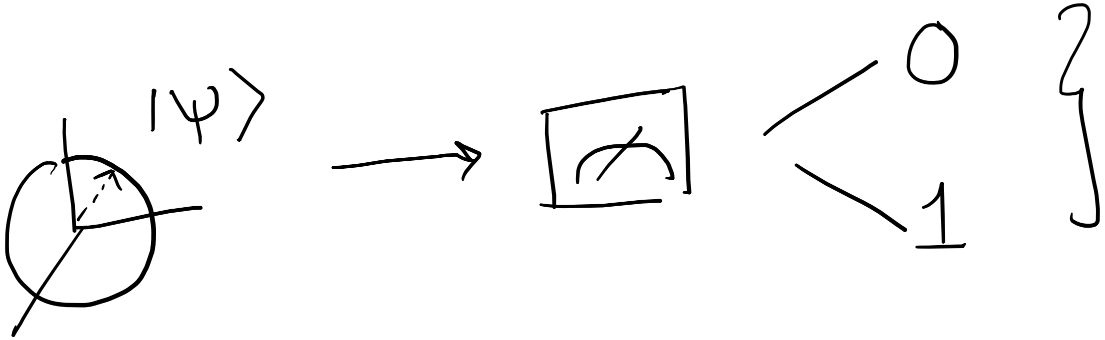
$|\psi\rangle$



Teleportation: Why?

- Classically: copying
- No cloning theorem
- Measurement looses the state

× A A ✓



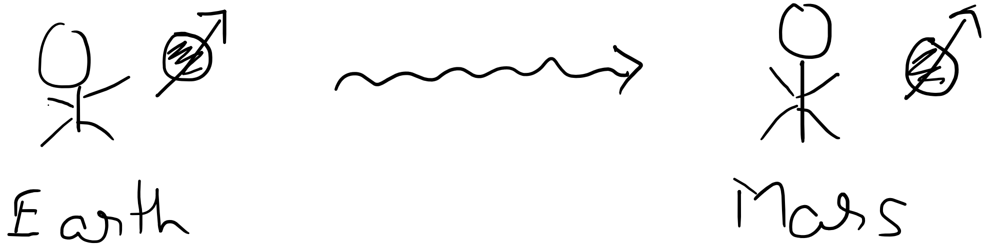
classical

Information : α, β !

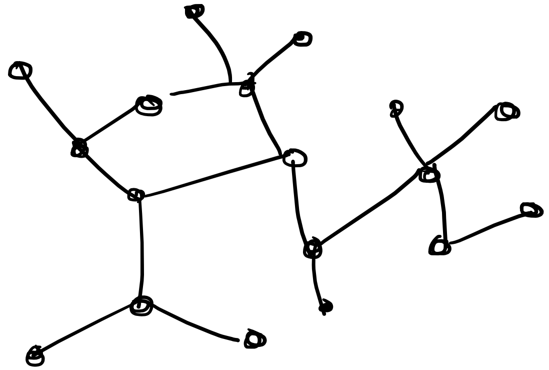
Teleportation: Applications

Quantum communications:

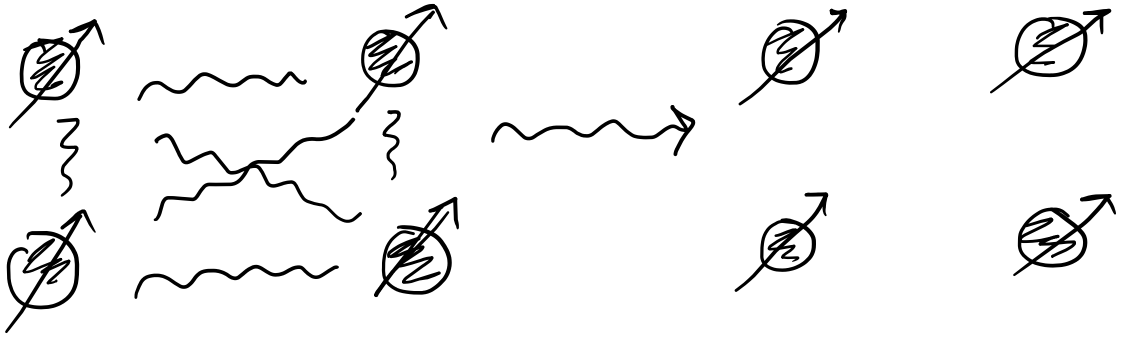
Direct quantum bit transfer is difficult



Quantum Networks



Distributed quantum computing



Thank you for your attention!

References

- C. Bennett, G. Brassard, C. Crépeau, R. Jozsa, A. Peres and W. Wootters, "Teleporting an unknown quantum state via dual classical and Einstein-Podolsky-Rosen channels", Physical Review Letters, vol. 70, no. 13, pp. 1895-1899, 1993. Available: 10.1103/physrevlett.70.1895.
- D. Bouwmeester, J. Pan, K. Mattle, M. Eibl, H. Weinfurter and A. Zeilinger, "Experimental quantum teleportation", Nature, vol. 390, no. 6660, pp. 575-579, 1997. Available: 10.1038/37539
- S. Pirandola, J. Eisert, C. Weedbrook, A. Furusawa and S. Braunstein, "Advances in quantum teleportation", Nature Photonics, vol. 9, no. 10, pp. 641-652, 2015. Available: 10.1038/nphoton.2015.154.
- T. Liu, "The Applications and Challenges of Quantum Teleportation", Journal of Physics: Conference Series, vol. 1634, p. 012089, 2020. Available: 10.1088/1742-6596/1634/1/012089.
- G. Milburn, "on “The Physics of Quantum Information: Quantum Cryptography, Quantum Teleportation, Quantum Computation" (edited by D Bouwmeester, A Ekert & A Zeilinger)", Quantum Information and Computation, vol. 1, no. 3, pp. 89-90, 2001. Available: 10.26421/qic1.3-9.
- Michael A. Nielsen and Isaac L. Chuang. 2011. Quantum Computation and Quantum Information: 10th Anniversary Edition (10th. ed.). Cambridge University Press, USA.